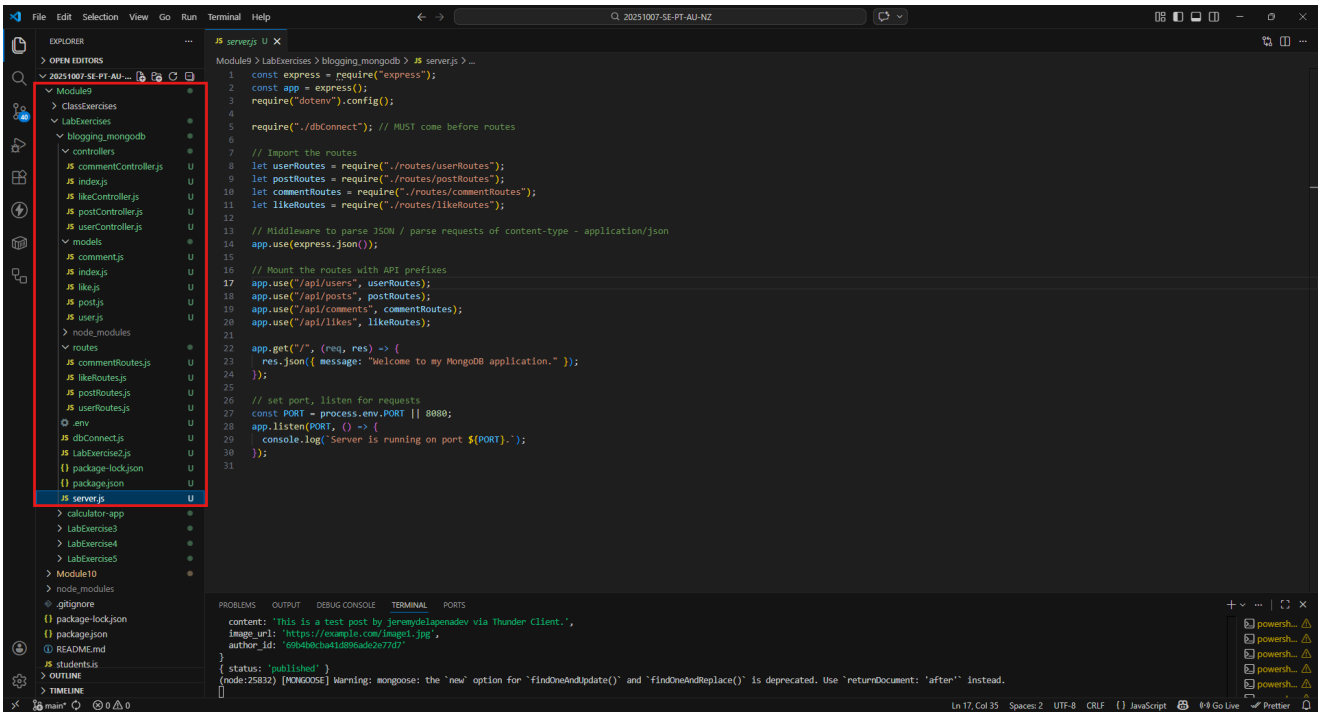
Module 9 Lab Exercise 2 - Backend Database via MongoDB

Module 9 - Lab Exercise 2 (Page 29)
https://drive.google.com/drive/folders/1Az2LS7OiuGDsx1MB7rPtXtk_ak_t6_6W

Create an express back-end application for a Blog website using MongoDB. You should refer to your database model from
Module 8 for this, ensuring that your app matches the model.

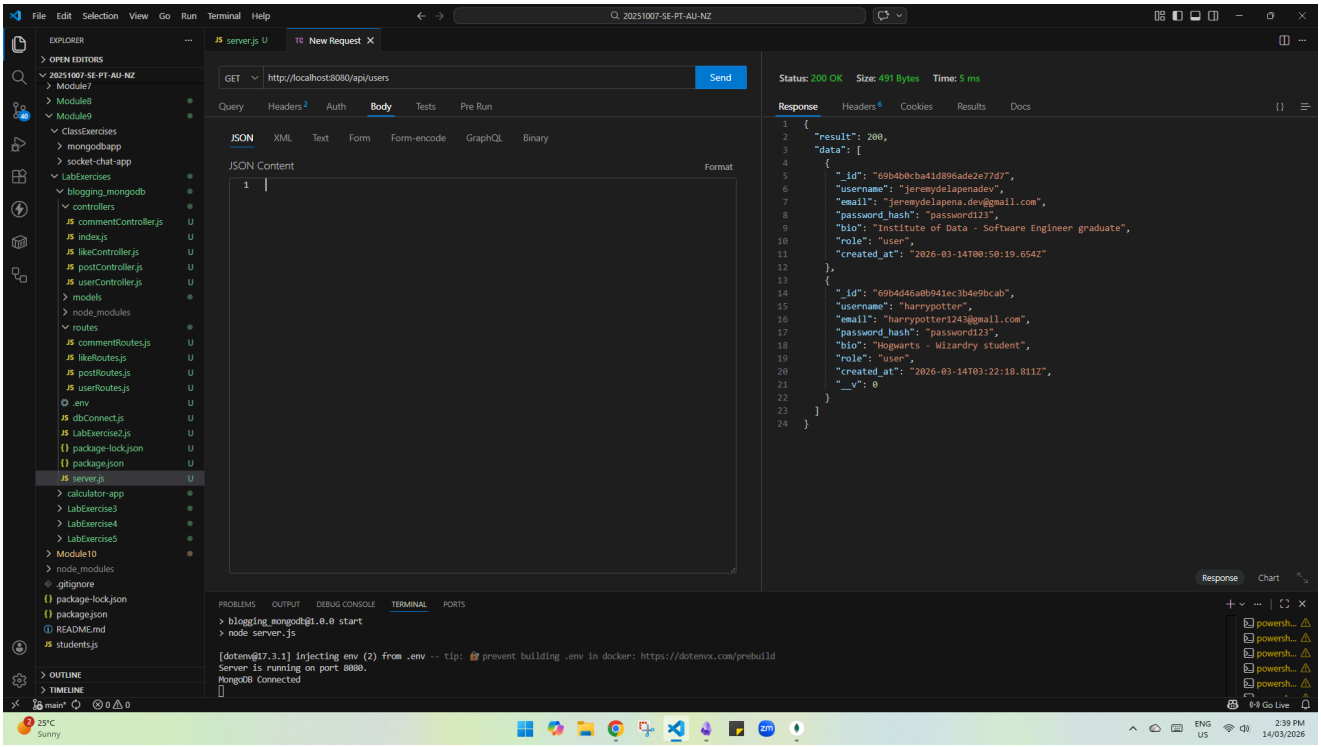
Requirements:
-Your system should have a proper MVC Structure
-The system should be able to create users.
-The users should be able to create multiple posts (posts should be very basic with title, description and image)
-Other users should be able to like the posts and comment on the posts.

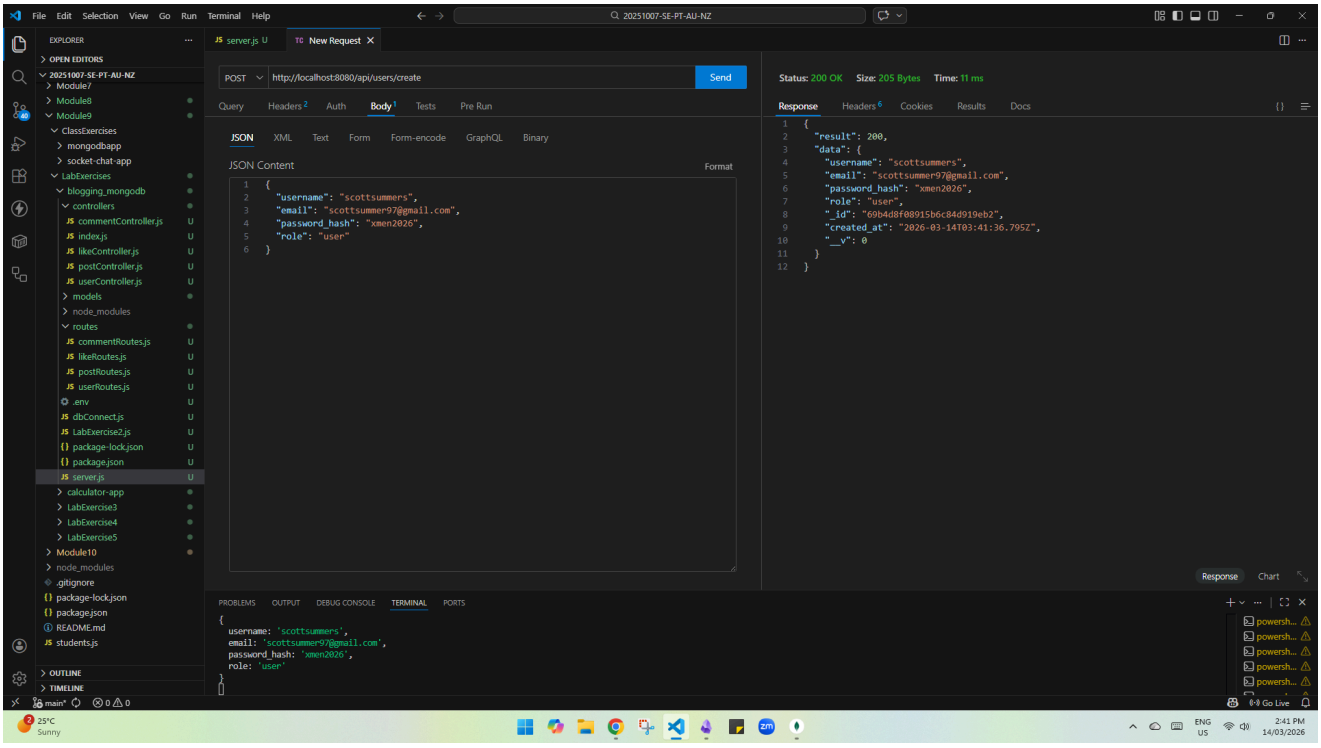
[/] Your system should have a proper MVC Structure

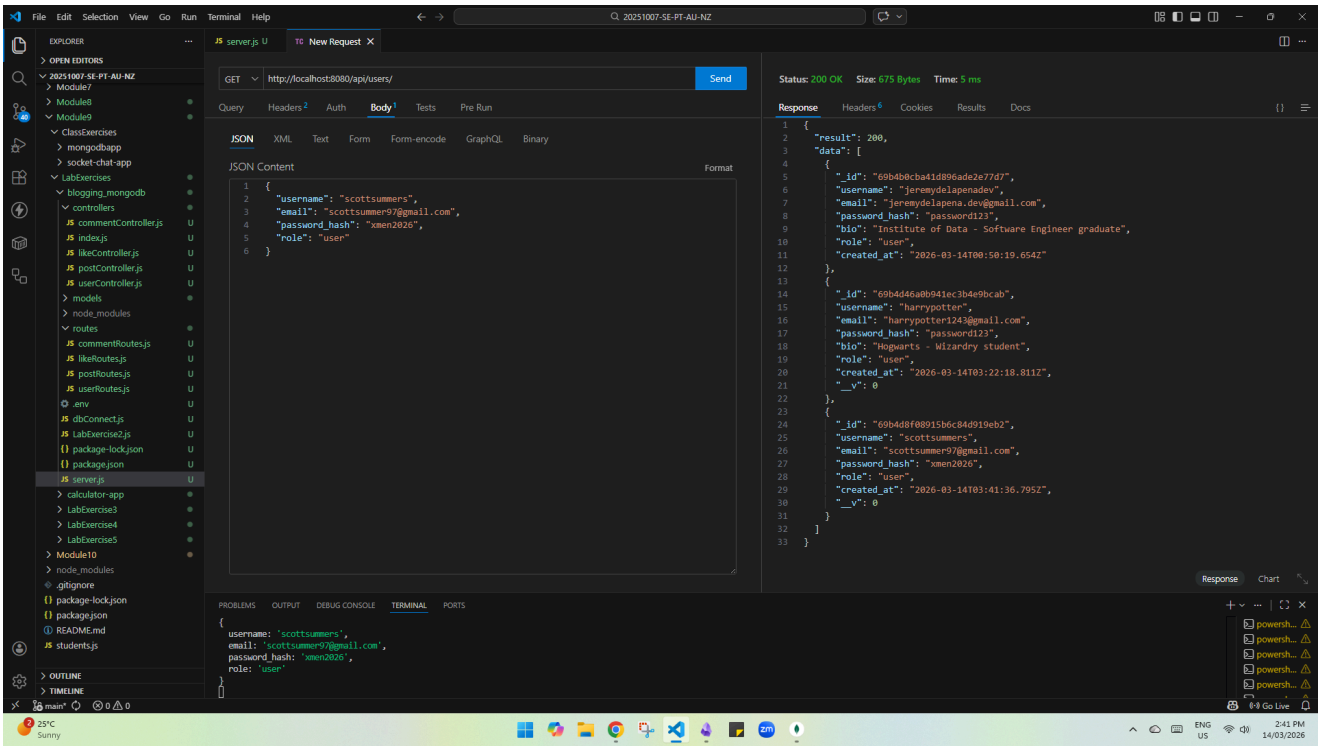


[/] The system should be able to create users.

1.) Creating a user.

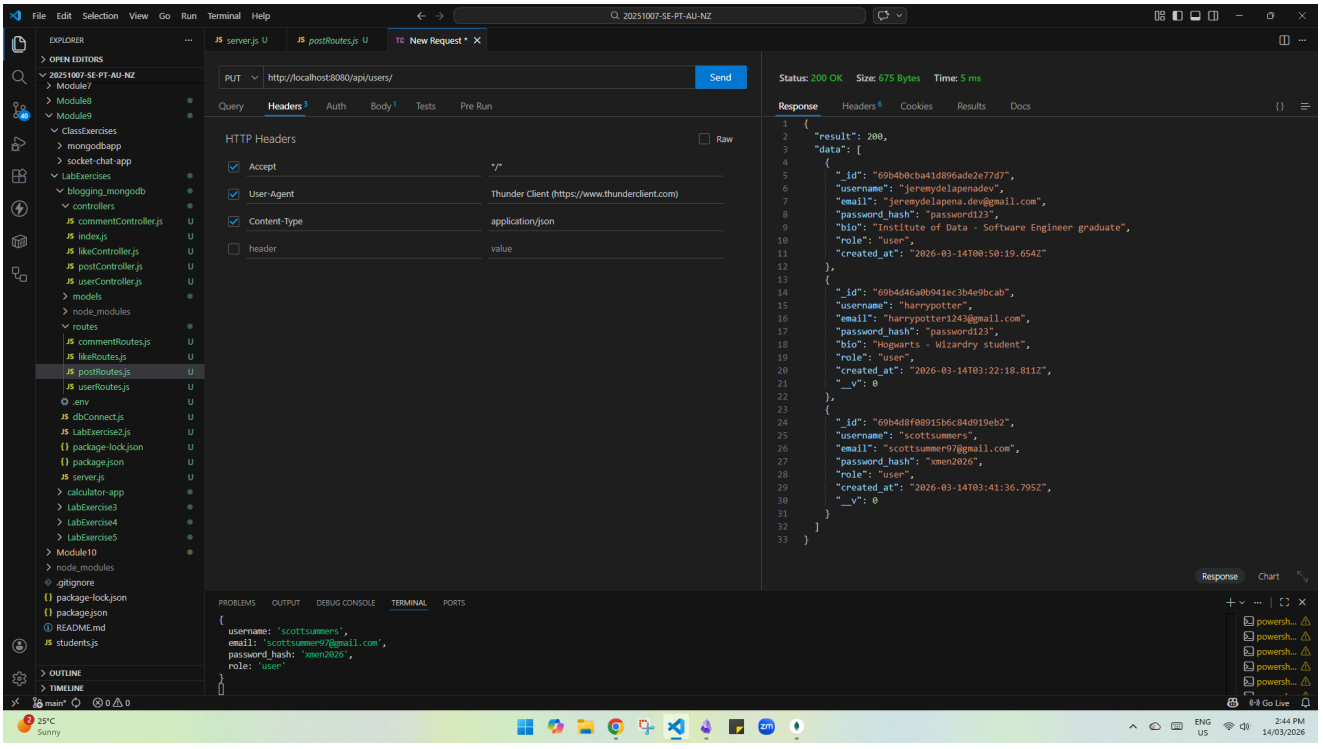


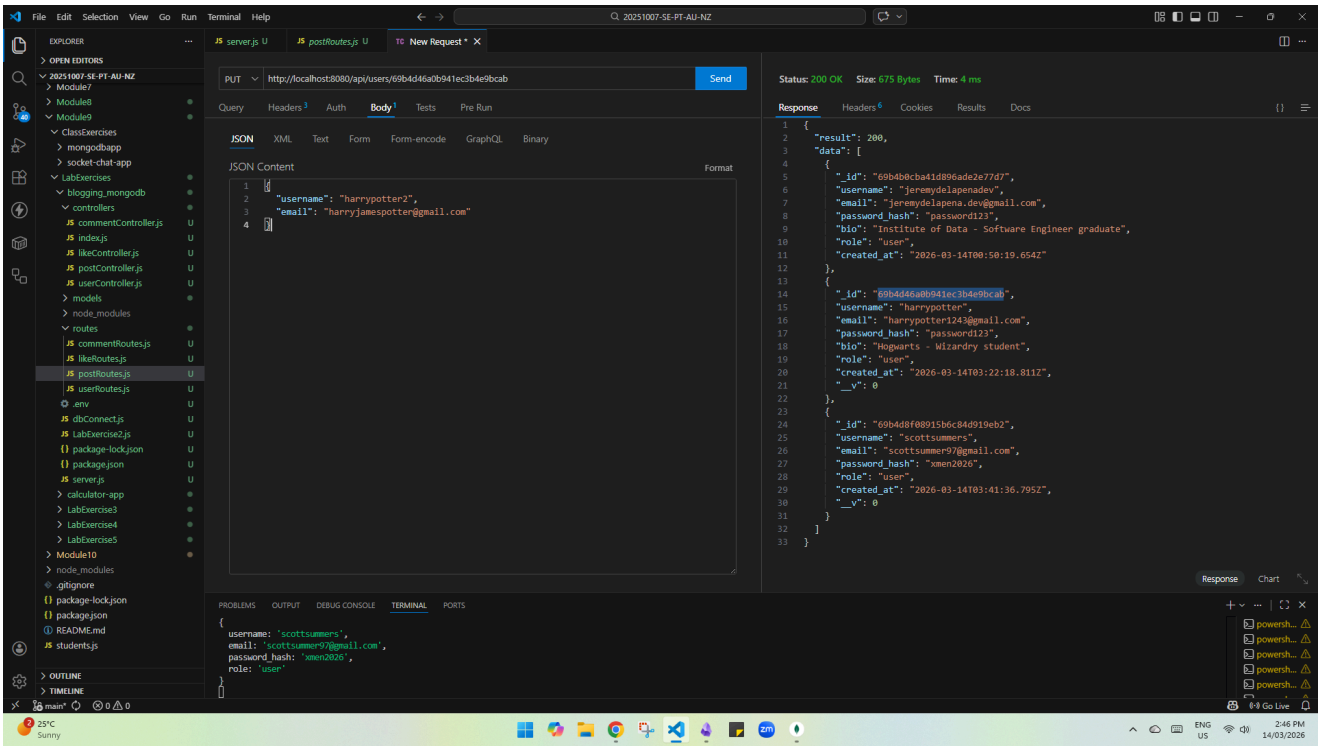


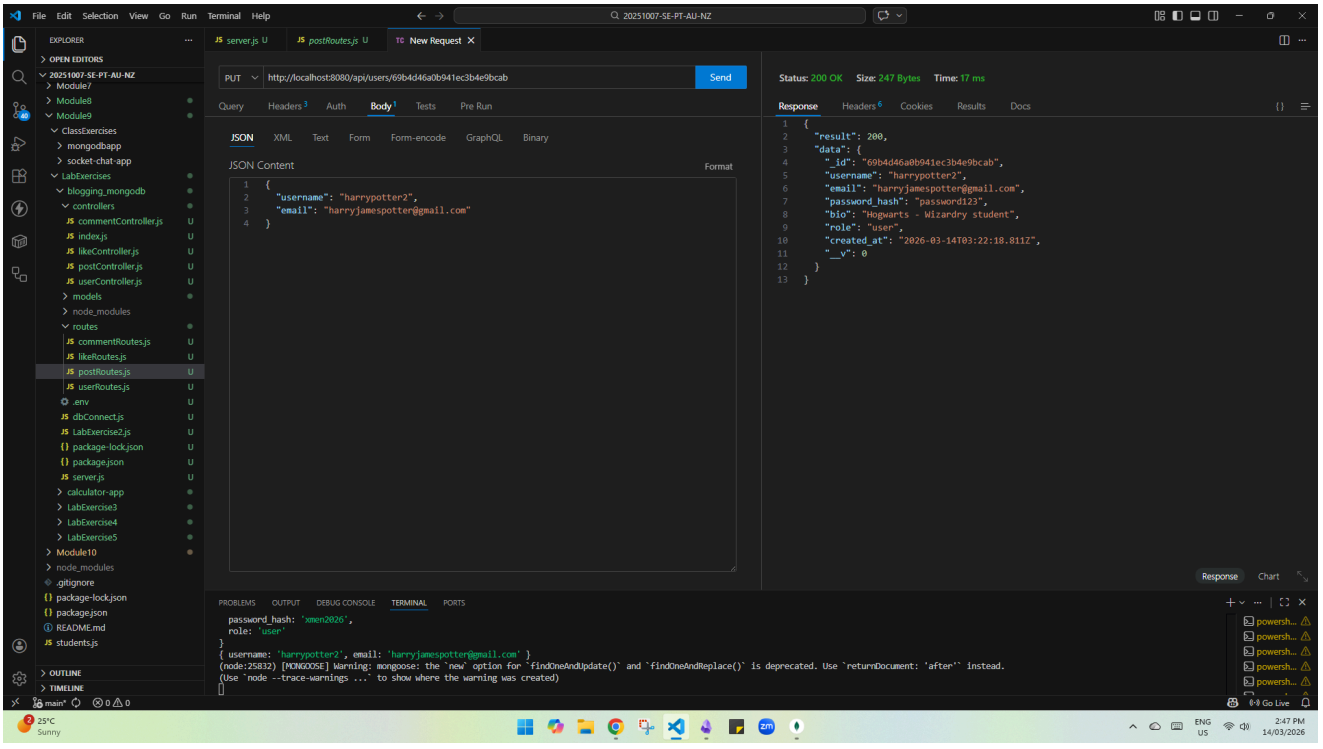


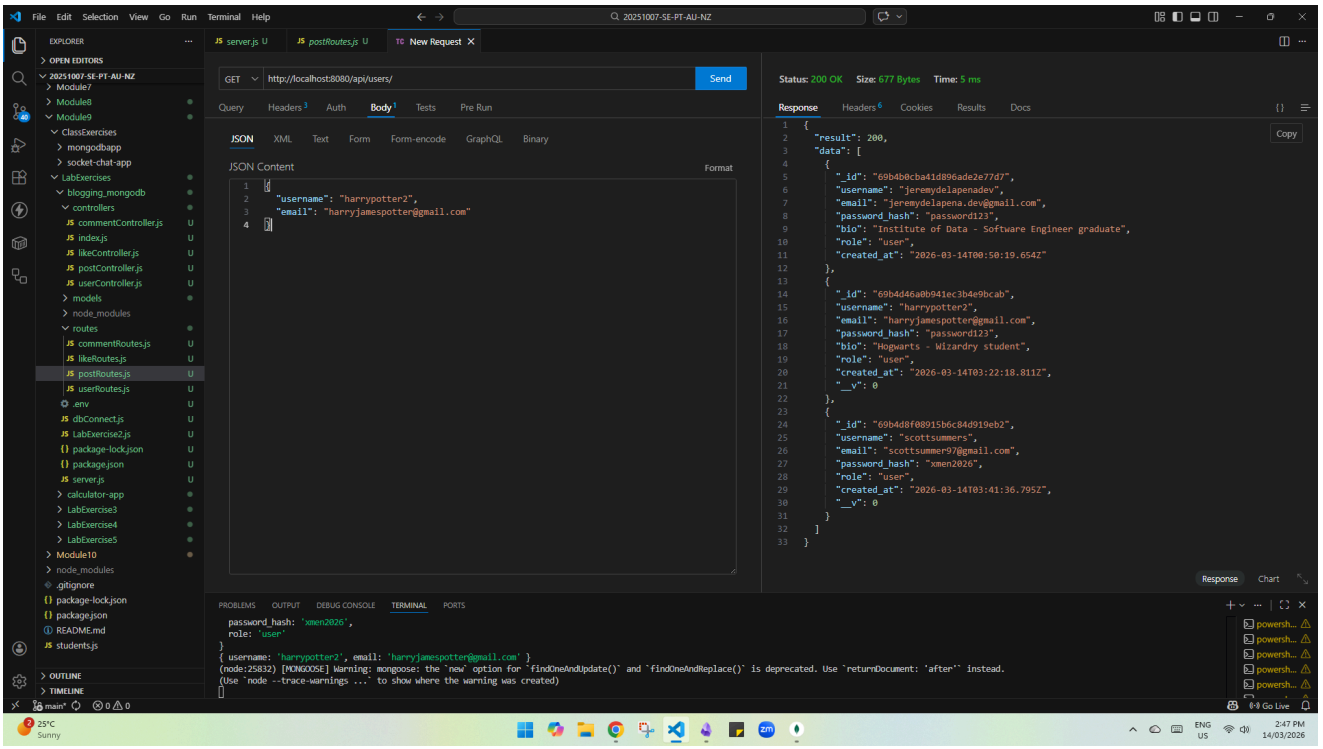
2.) Updating a user.

Note: Added `Content-Type` and `application/json` before putting data.

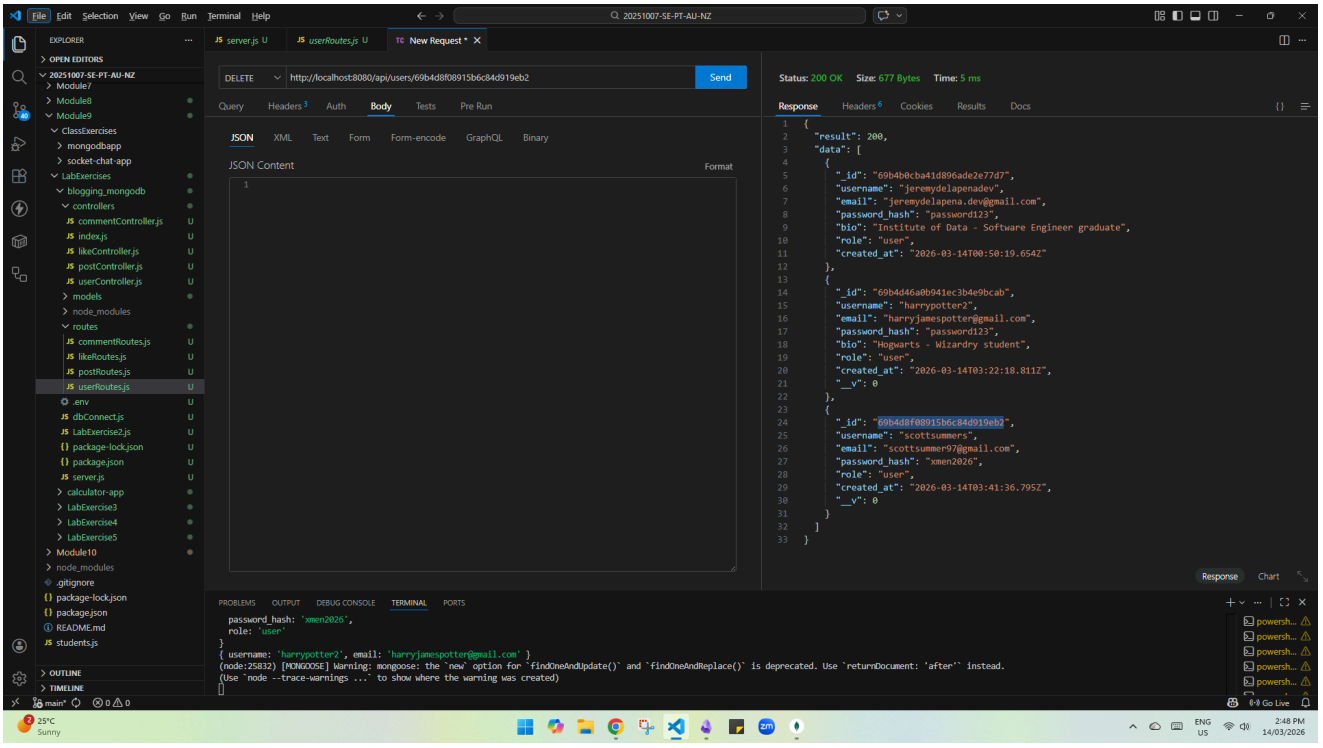


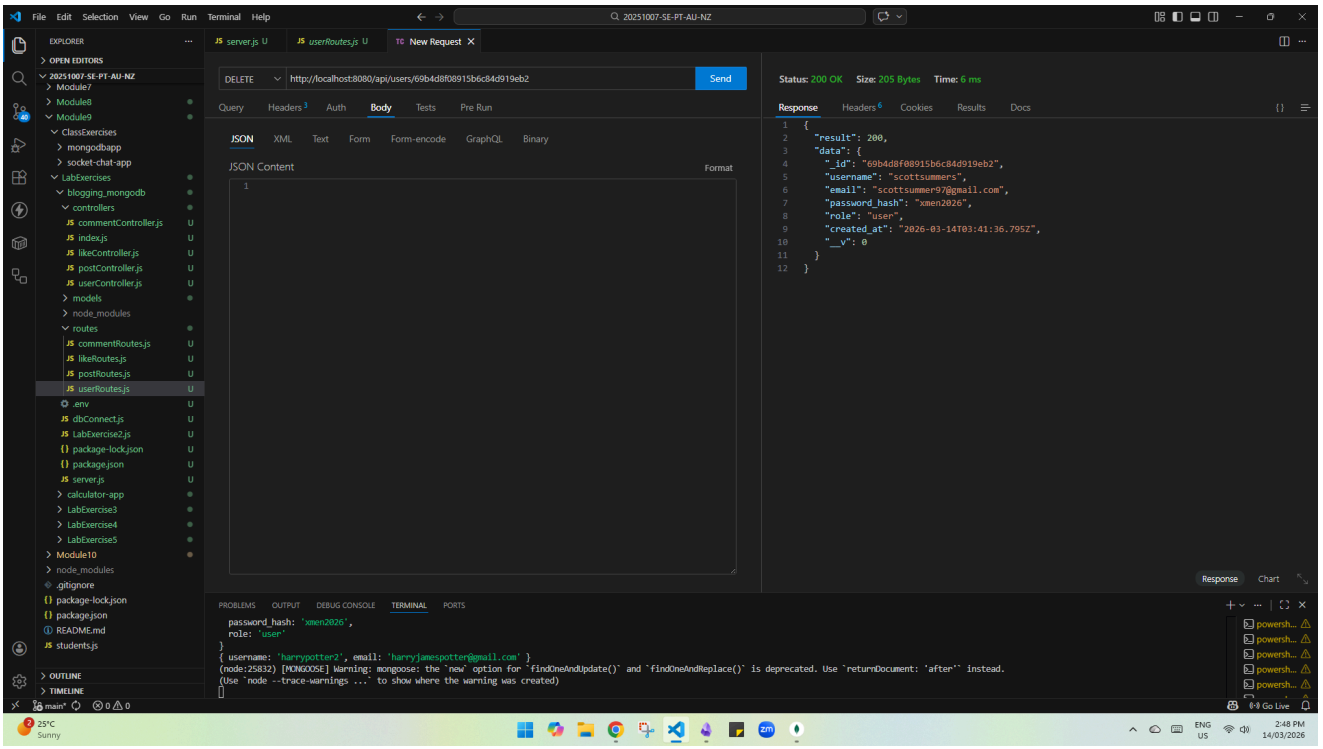

PUT
http://localhost:8080/api/users/69b4d46a0b941ec3b4e9bcab
Send
Status: 200 OK
Size: 675 Bytes
Time: 4 ms
{
"username": "harrypotter2",
"email": "harryjamespotter@gmail.com"
}
{
"result": 200,
"data": [
{
"_id": "69b4b0cba41d896ade2e77d7",
"username": "jeremydelapenadev",
"email": "jeremydelapena.dev@gmail.com",
"password_hash": "password123",
"bio": "Institute of Data - Software Engineer graduate",
"role": "user",
"created_at": "2026-03-14T00:50:19.654Z"
},
{
"_id": "69b4d46a0b941ec3b4e9bcab",
"username": "harrypotter",
"email": "harrypotter1243@gmail.com",
"password_hash": "password123",
"bio": "Hogwarts - Wizardry student",
"role": "user",
"created_at": "2026-03-14T03:22:18.811Z",
"__v": 0
},
{
"_id": "69b4d8f08915b6c84d919eb2",
"username": "scottsummers",
"email": "scottsummer97@gmail.com",
"password_hash": "xmen2026",
"role": "user",
"created_at": "2026-03-14T03:41:36.795Z",
"__v": 0
}
]
}
{
username: 'scottsummers',
email: 'scottsummer97@gmail.com',
password_hash: 'xmen2026',
role: 'user'
}

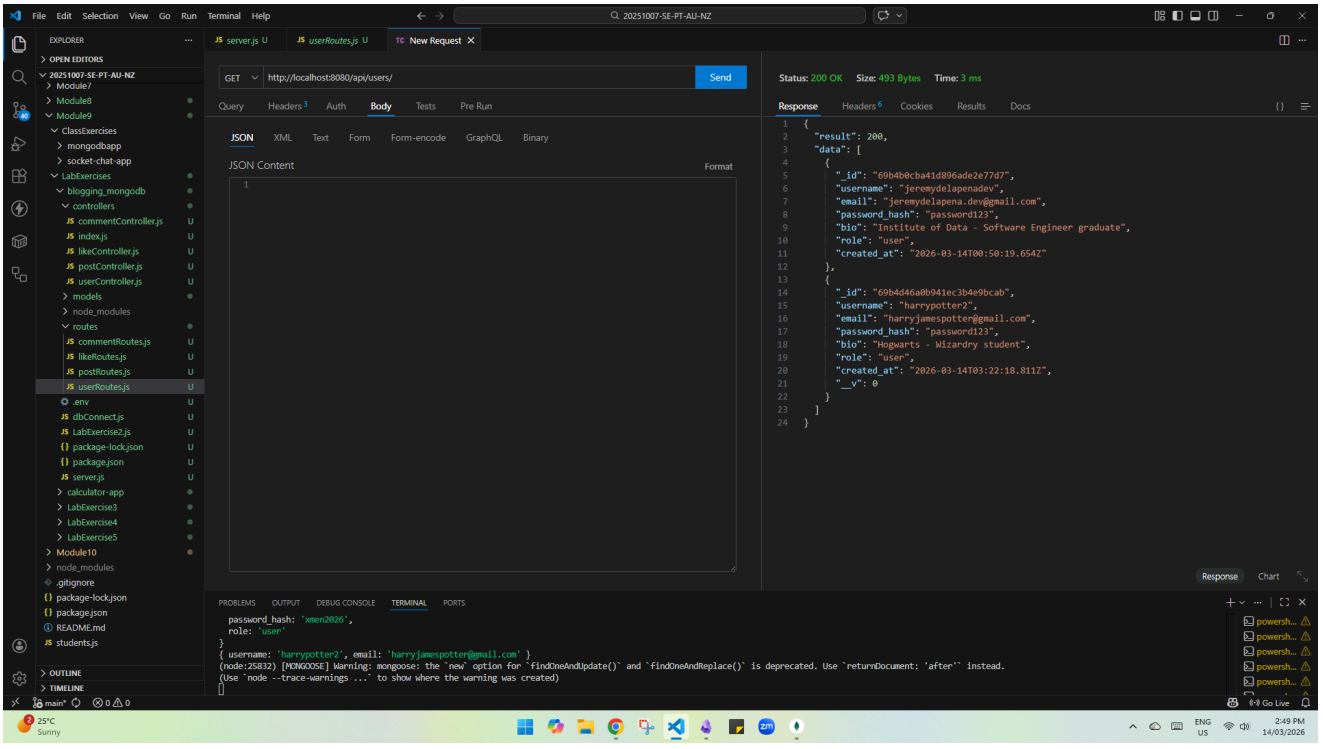

PUT
http://localhost:8080/api/users/69b4d46a0b941ec3b4e9bcab
Send
Status: 200 OK
Size: 247 Bytes
Time: 17 ms
{
"username": "harrypotter2",
"email": "harryjamespotter@gmail.com"
}
{
"result": 200,
"data": {
"_id": "69b4d46a0b941ec3b4e9bcab",
"username": "harrypotter2",
"email": "harryjamespotter@gmail.com",
"password_hash": "password123",
"bio": "Hogwarts - Wizardry student",
"role": "user",
"created_at": "2026-03-14T03:22:18.811Z",
"__v": 0
}
}
password_hash: 'xmen2026',
role: 'user'
}
{ username: 'harrypotter2', email: 'harryjamespotter@gmail.com' }
(node:25832) [MONGOOSE] Warning: mongoose: the `new` option for `findOneAndUpdate()` and `findOneAndReplace()` is deprecated. Use `returnDocument: 'after'` instead.
(Use `node --trace-warnings ...` to show where the warning was created)



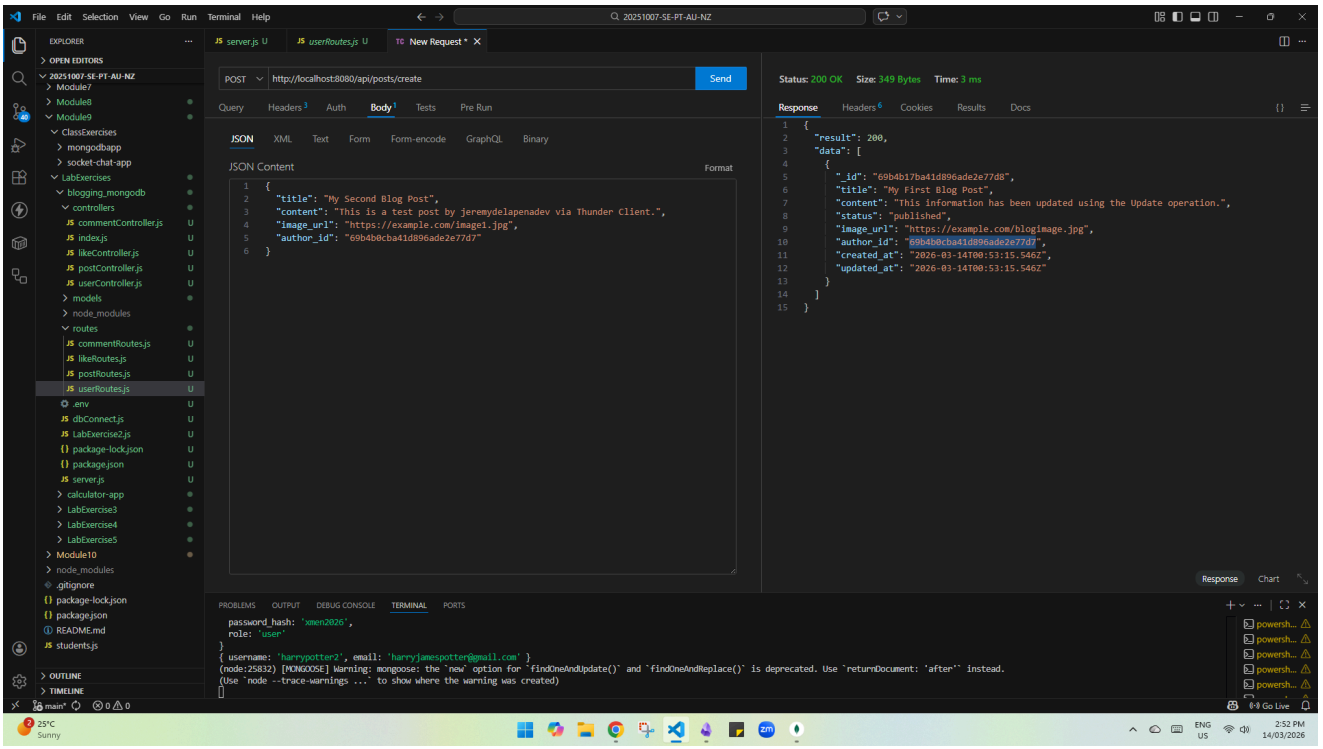
3.) Deleting a user

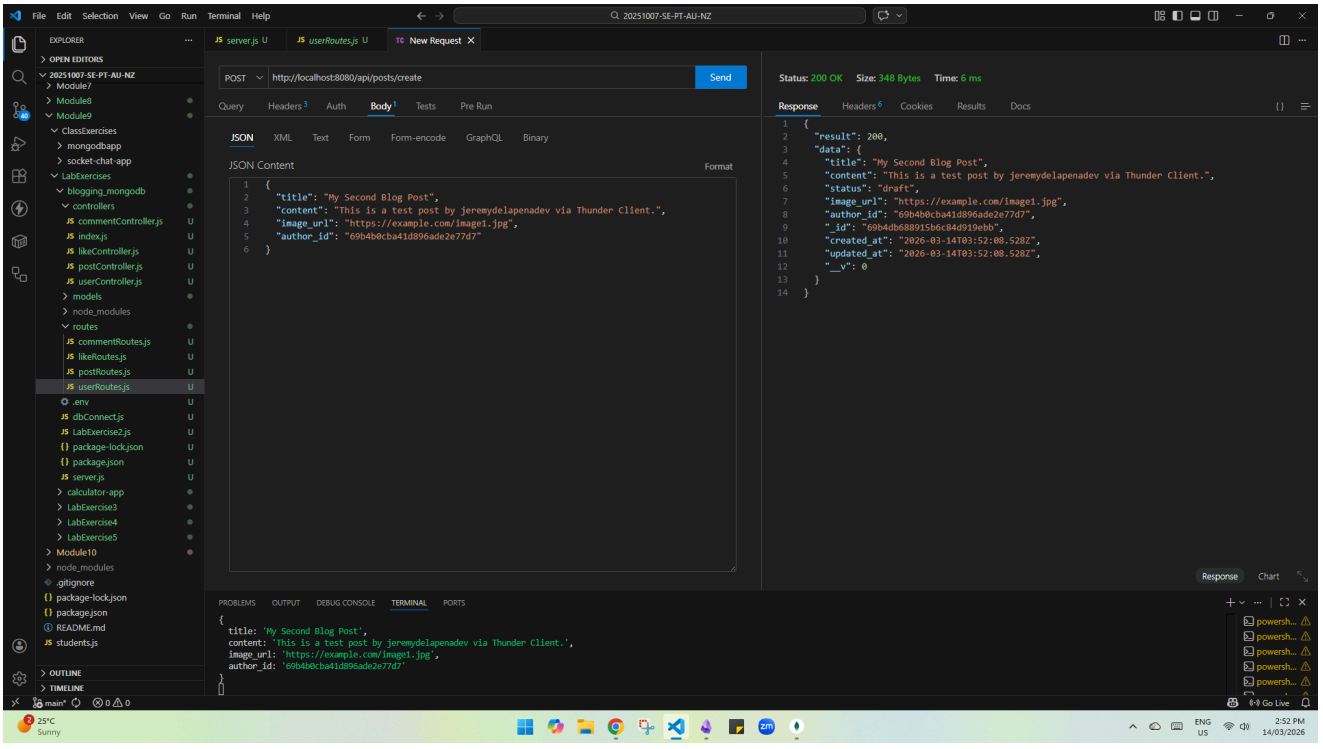




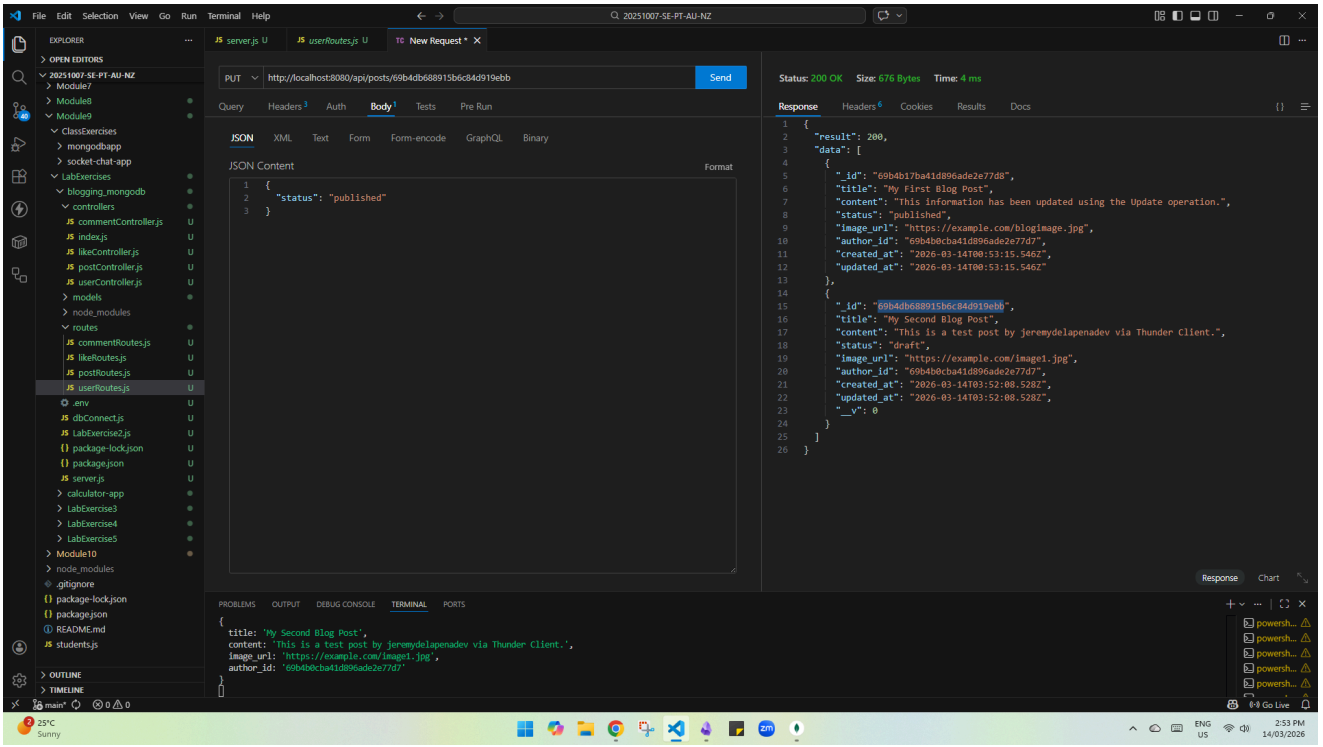


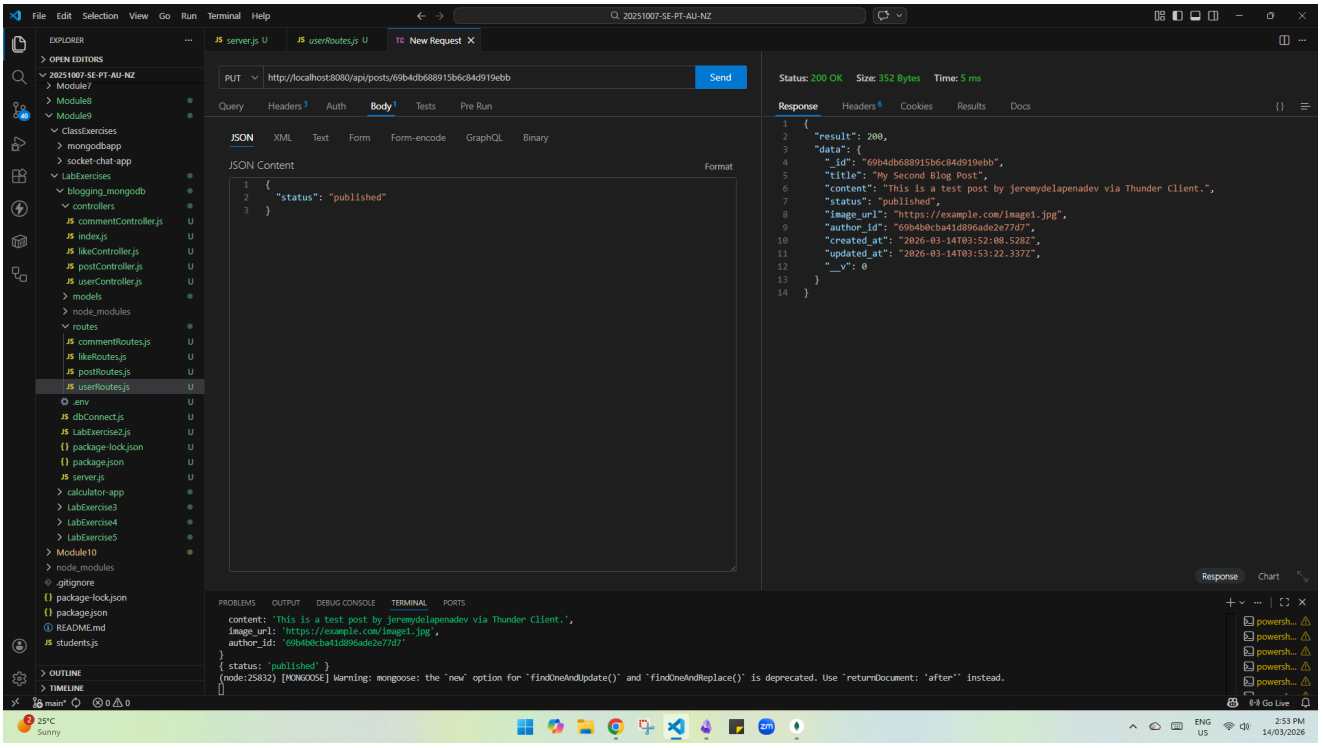
[/] The users should be able to create multiple posts (posts should be very basic with title, description and image)


File Edit Selection View Go Run Terminal Help
20251007-SE-PT-AU-NZ
server.js U
userRoutes.js U
New Request *
POST
http://localhost:8080/api/posts/create
Send
Query Headers 3 Auth Body 1 Tests Pre Run
JSON XML Text Form Form-encode GraphQL Binary
JSON Content
Format
{
"title": "My Second Blog Post",
"content": "This is a test post by jeremydelapenadev via Thunder Client.",
"image_url": "https://example.com/image1.jpg",
"author_id": "69b4b0cba41d896ade2e77d7"
}
Status: 200 OK Size: 349 Bytes Time: 3 ms
Response Headers 6 Cookies Results Docs
{
"result": 200,
"data": [
{
"_id": "69b4b17ba41d896ade2e77d8",
"title": "My First Blog Post",
"content": "This information has been updated using the Update operation.",
"status": "published",
"image_url": "https://example.com/blogimage.jpg",
"author_id": "69b4b0cba41d896ade2e77d7",
"created_at": "2026-03-14T00:53:15.546Z",
"updated_at": "2026-03-14T00:53:15.546Z"
}
]
}
Response Chart
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
password_hash: 'xmen2026',
role: 'user'
}
{ username: 'harrypotter2', email: 'harryjamespotter@gmail.com' }
(node:25832) [MONGOOSE] Warning: mongoose: the `new` option for `findOneAndUpdate()` and `findOneAndReplace()` is deprecated. Use `returnDocument: 'after'` instead.
(Use `node --trace-warnings ...` to show where the warning was created)
OUTLINE
TIMELINE
main*
Go Live
25°C Sunny
ENG US
2:52 PM 14/03/2026


File Edit Selection View Go Run Terminal Help
20251007-SE-PT-AU-NZ
server.js U
userRoutes.js U
New Request
POST
http://localhost:8080/api/posts/create
Send
Query Headers 3 Auth Body 1 Tests Pre Run
JSON XML Text Form Form-encode GraphQL Binary
JSON Content
Format
{
"title": "My Second Blog Post",
"content": "This is a test post by jeremydelapenadev via Thunder Client.",
"image_url": "https://example.com/image1.jpg",
"author_id": "69b4b0cba41d896ade2e77d7"
}
Status: 200 OK Size: 348 Bytes Time: 6 ms
Response Headers 6 Cookies Results Docs
{
"result": 200,
"data": {
"title": "My Second Blog Post",
"content": "This is a test post by jeremydelapenadev via Thunder Client.",
"status": "draft",
"image_url": "https://example.com/image1.jpg",
"author_id": "69b4b0cba41d896ade2e77d7",
"_id": "69b4db688915b6c84d919ebb",
"created_at": "2026-03-14T03:52:08.528Z",
"updated_at": "2026-03-14T03:52:08.528Z",
"__v": 0
}
}
Response Chart
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
{
title: 'My Second Blog Post',
content: 'This is a test post by jeremydelapenadev via Thunder Client.',
image_url: 'https://example.com/image1.jpg',
author_id: '69b4b0cba41d896ade2e77d7'
}
OUTLINE
TIMELINE
main*
Go Live
25°C Sunny
ENG US
2:52 PM 14/03/2026

Additional: Updating the `status` from `draft` to `published`

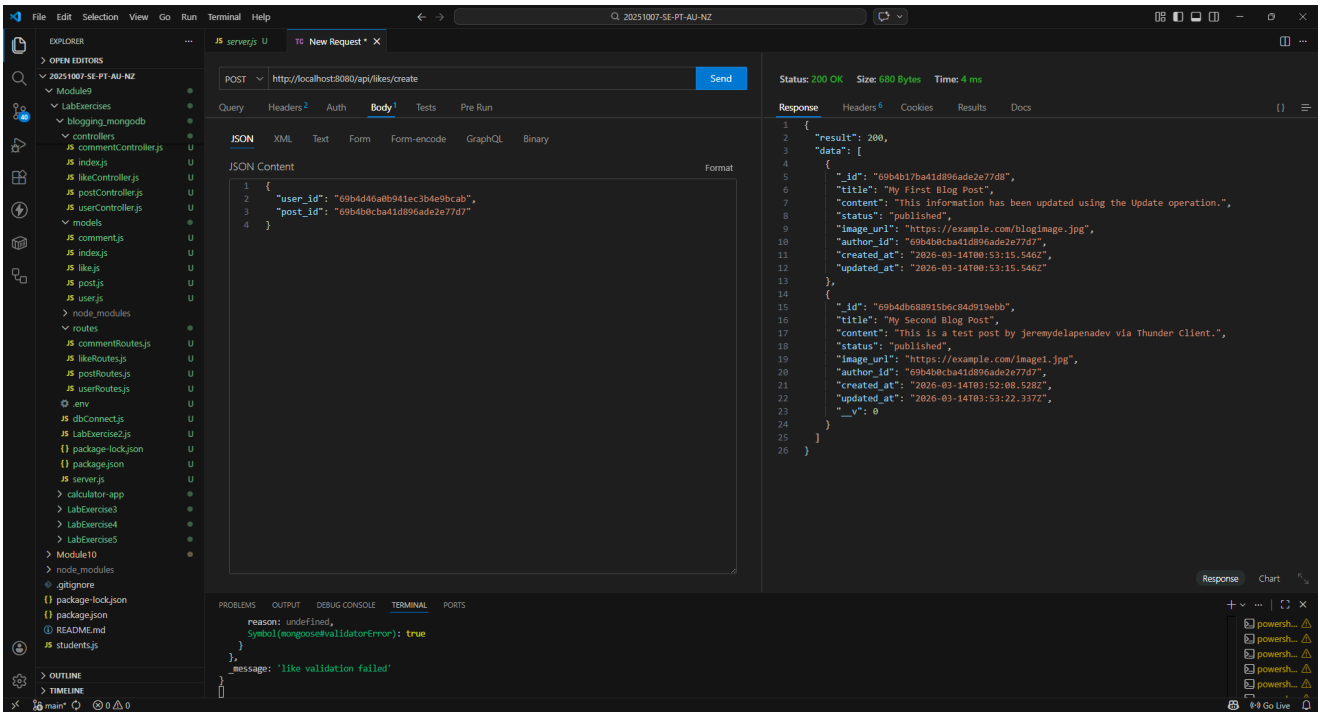


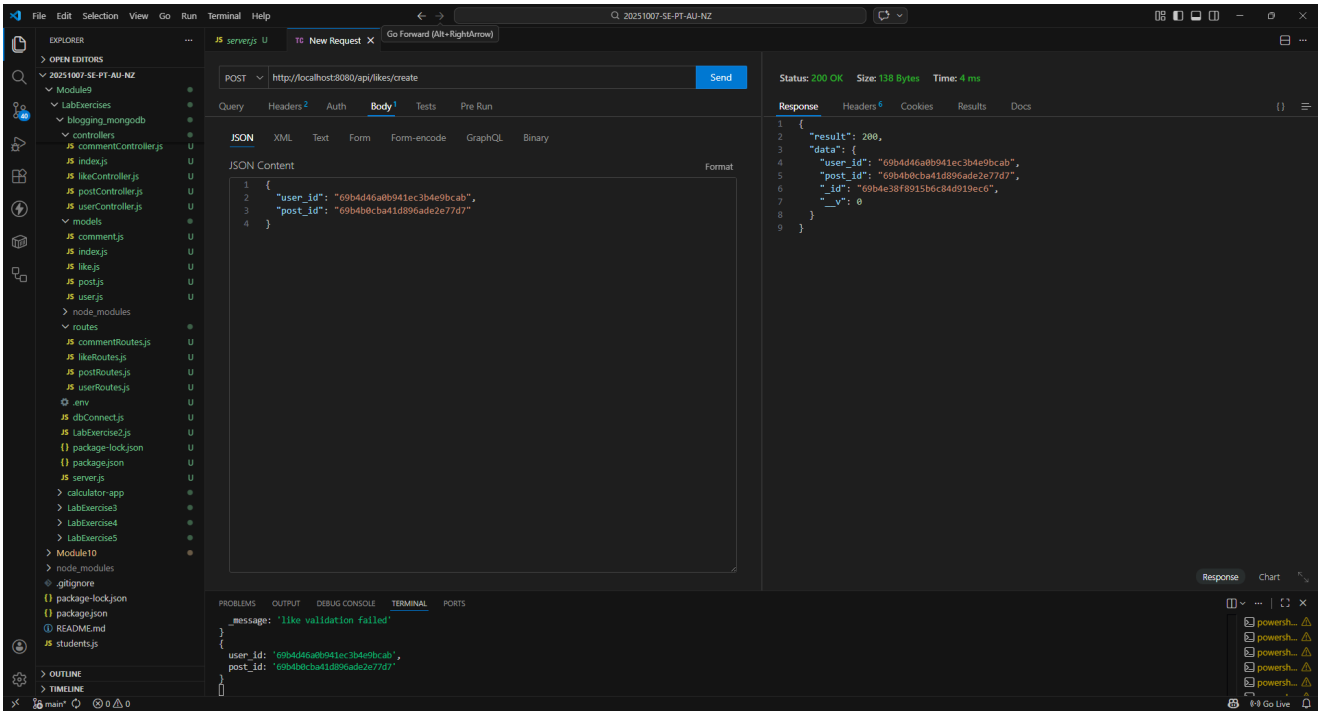


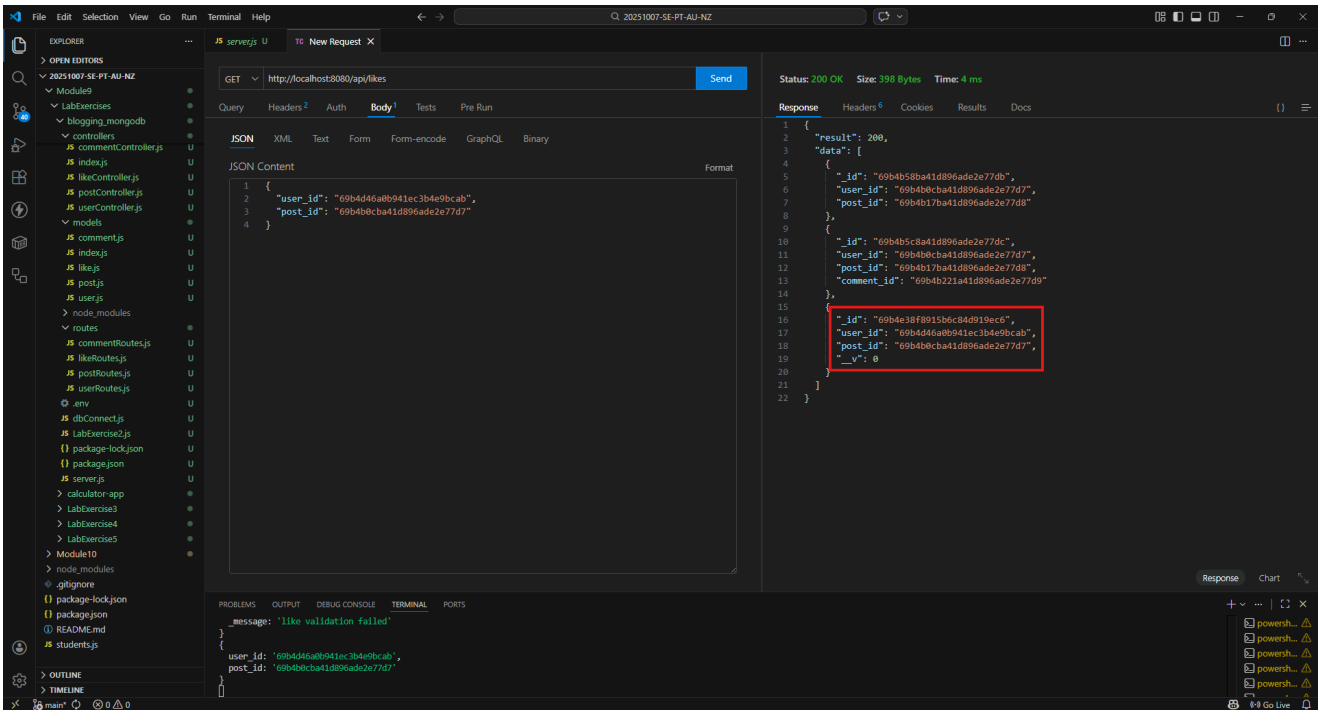
[/] Other users should be able to like the posts.

`user_id` of `jeremydelapenadev` : 69b4b0cba41d896ade2e77d7
`user_id` of `harrypotter2` : 69b4d46a0b941ec3b4e9bcab

`post_id` of `jeremydelapenadev` : 69b4b17ba41d896ade2e77d8 (First Blog Post)
`post_id` of `jeremydelapenadev` : 69b4db688915b6c84d919ebb (Second Blog Post)


File Edit Selection View Go Run Terminal Help
20251007-SE-PT-AU-NZ
EXPLORER
OPEN EDITORS
20251007-SE-PT-AU-NZ
Module9
LabExercises
blogging_mongodb
controllers
commentController.js
index.js
likeController.js
postController.js
userController.js
models
comment.js
index.js
like.js
post.js
user.js
node_modules
routes
commentRoutes.js
likeRoutes.js
postRoutes.js
userRoutes.js
.env
dbConnect.js
LabExercise2.js
package-lock.json
package.json
server.js
calculator-app
LabExercise3
LabExercise4
LabExercise5
Module10
node_modules
.gitignore
package-lock.json
package.json
README.md
students.js
OUTLINE
TIMELINE
server.js U
New Request *
POST
http://localhost:8080/api/likes/create
Send
Query
Headers 2
Auth
Body 1
Tests
Pre Run
JSON
XML
Text
Form
Form-encode
GraphQL
Binary
JSON Content
Format
{
"user_id": "69b4d46a0b941ec3b4e9bcab",
"post_id": "69b4b0cba41d896ade2e77d7"
}
Status: 200 OK
Size: 680 Bytes
Time: 4 ms
Response
Headers 6
Cookies
Results
Docs
{
"result": 200,
"data": [
{
"_id": "69b4b17ba41d896ade2e77d8",
"title": "My First Blog Post",
"content": "This information has been updated using the Update operation.",
"status": "published",
"image_url": "https://example.com/blogimage.jpg",
"author_id": "69b4b0cba41d896ade2e77d7",
"created_at": "2026-03-14T00:53:15.546Z",
"updated_at": "2026-03-14T00:53:15.546Z"
},
{
"_id": "69b4db688915b6c84d919ebb",
"title": "My Second Blog Post",
"content": "This is a test post by jeremydelapenadev via Thunder Client.",
"status": "published",
"image_url": "https://example.com/image1.jpg",
"author_id": "69b4b0cba41d896ade2e77d7",
"created_at": "2026-03-14T03:52:08.528Z",
"updated_at": "2026-03-14T03:53:22.337Z",
"__v": 0
}
]
}
Response
Chart
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
reason: undefined,
Symbol(mongoose#validatorError): true
},
_message: 'like validation failed'
}
main*
Go Live

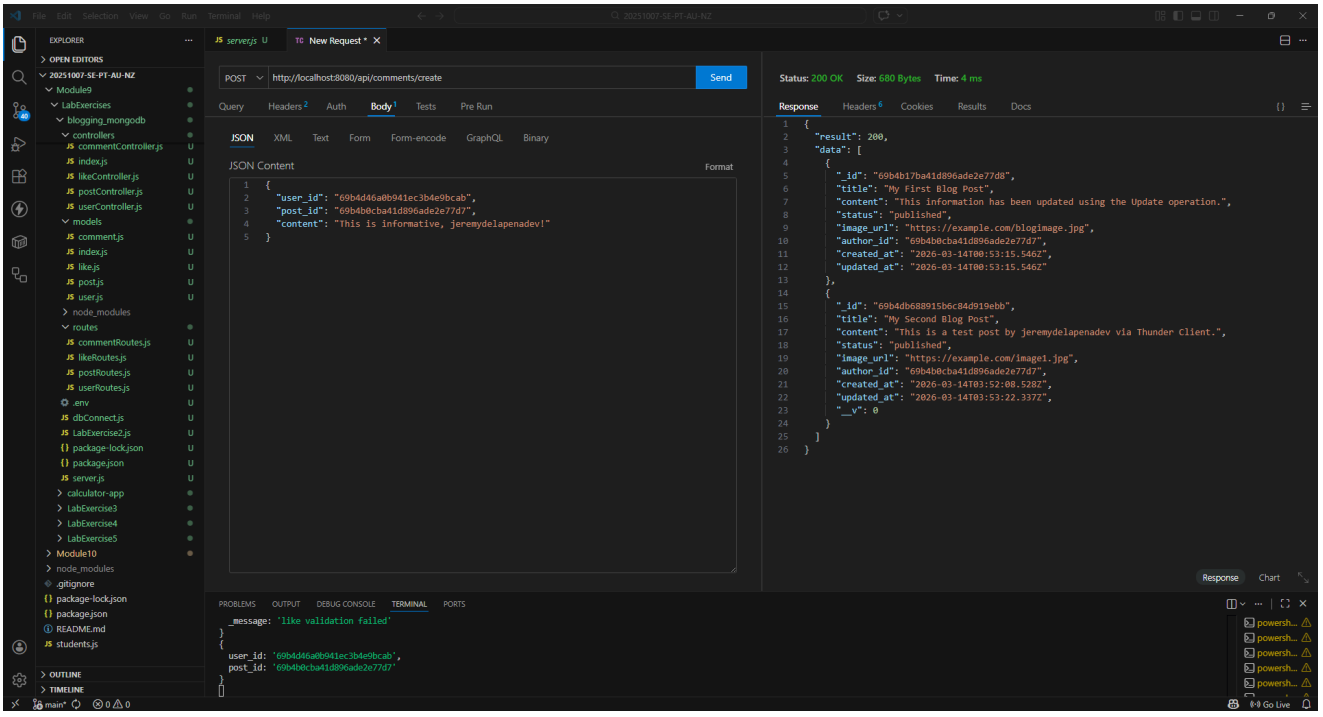

File Edit Selection View Go Run Terminal Help
20251007-SE-PT-AU-NZ
Go Forward (Alt+RightArrow)
server.js U
New Request
POST
http://localhost:8080/api/likes/create
Send
Query
Headers 2
Auth
Body 1
Tests
Pre Run
JSON
XML
Text
Form
Form-encode
GraphQL
Binary
JSON Content
Format
{
"user_id": "69b4d46a0b941ec3b4e9bcab",
"post_id": "69b4b0cba41d896ade2e77d7"
}
Status: 200 OK
Size: 138 Bytes
Time: 4 ms
Response
Headers 6
Cookies
Results
Docs
{
"result": 200,
"data": {
"user_id": "69b4d46a0b941ec3b4e9bcab",
"post_id": "69b4b0cba41d896ade2e77d7",
"_id": "69b4e38f8915b6c84d919ec6",
"__v": 0
}
}
Response
Chart
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
_message: 'like validation failed'
}
{
user_id: '69b4d46a0b941ec3b4e9bcab',
post_id: '69b4b0cba41d896ade2e77d7'
}
main*
Go Live

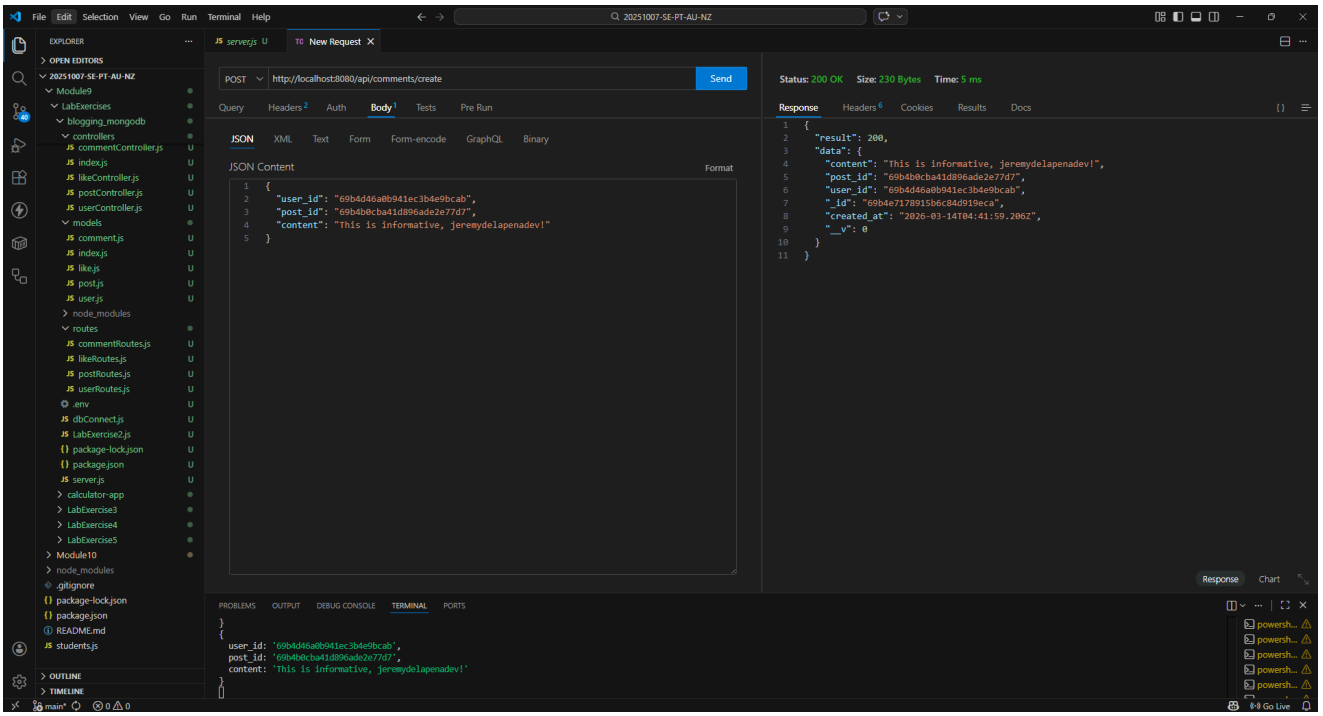


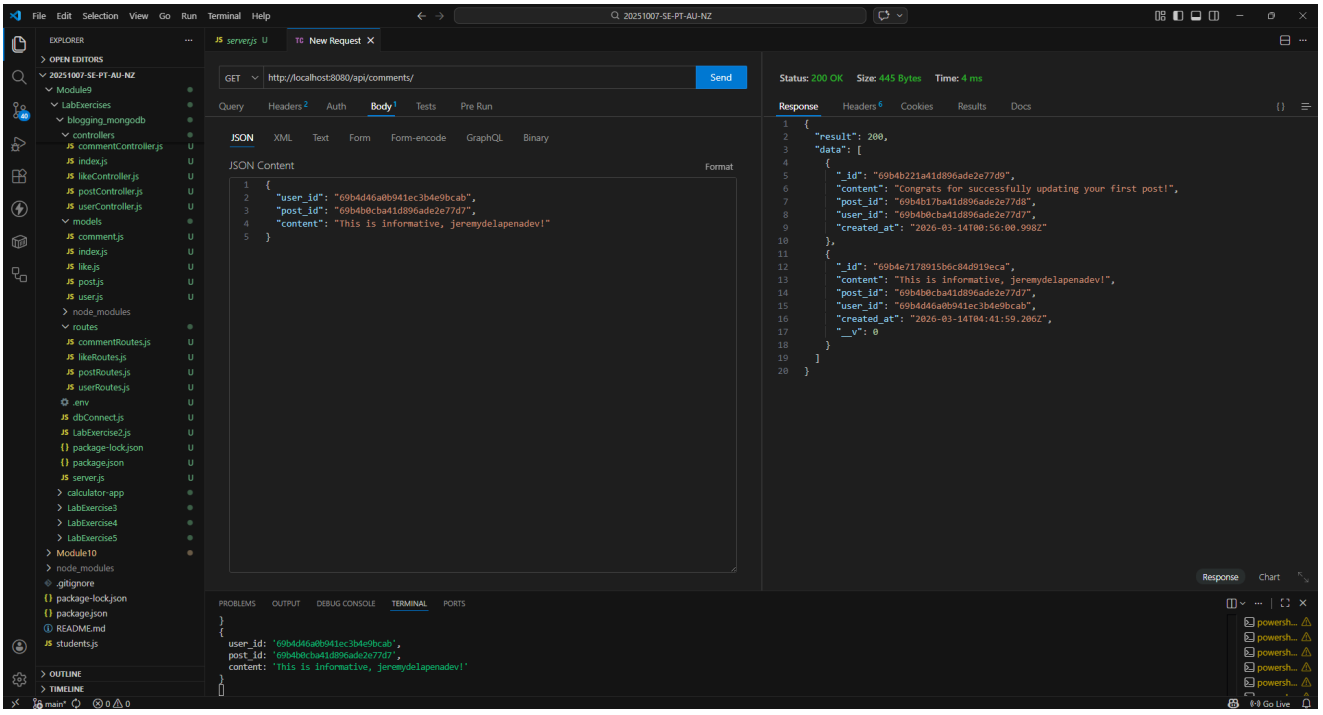
[/] Other users should be able to comment on the posts.

`user_id` of `jeremydelapenadev` : 69b4b0cba41d896ade2e77d7
`user_id` of `harrypotter2` : 69b4d46a0b941ec3b4e9bcab

`post_id` of `jeremydelapenadev` : 69b4b17ba41d896ade2e77d8 (First Blog Post)
`post_id` of `jeremydelapenadev` : 69b4db688915b6c84d919ebb (Second Blog Post)




File Edit Selection View Go Run Terminal Help
20251007-SE-PT-AU-NZ
EXPLORER
OPEN EDITORS
20251007-SE-PT-AU-NZ
Module9
LabExercises
blogging_mongodb
controllers
commentController.js
index.js
likeController.js
postController.js
userController.js
models
comment.js
index.js
like.js
post.js
user.js
node_modules
routes
commentRoutes.js
likeRoutes.js
postRoutes.js
userRoutes.js
.env
dbConnect.js
LabExercise2.js
package-lock.json
package.json
server.js
calculator-app
LabExercise3
LabExercise4
LabExercise5
Module10
node_modules
.gitignore
package-lock.json
package.json
README.md
students.js
OUTLINE
TIMELINE
server.js U
New Request
POST
http://localhost:8080/api/comments/create
Send
Query
Headers 2
Auth
Body 1
Tests
Pre Run
JSON
XML
Text
Form
Form-encode
GraphQL
Binary
JSON Content
Format
{
"user_id": "69b4d46a0b941ec3b4e9bcab",
"post_id": "69b4b0cba41d896ade2e77d7",
"content": "This is informative, jeremydelapenadev!"
}
Status: 200 OK
Size: 230 Bytes
Time: 5 ms
Response
Headers 6
Cookies
Results
Docs
{
"result": 200,
"data": {
"content": "This is informative, jeremydelapenadev!",
"post_id": "69b4b0cba41d896ade2e77d7",
"user_id": "69b4d46a0b941ec3b4e9bcab",
"_id": "69b4e7178915b6c84d919eca",
"created_at": "2026-03-14T04:41:59.206Z",
"__v": 0
}
}
Response
Chart
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
}
{
user_id: '69b4d46a0b941ec3b4e9bcab',
post_id: '69b4b0cba41d896ade2e77d7',
content: 'This is informative, jeremydelapenadev!'
}
powersh...
main*
0 0
Go Live


File Edit Selection View Go Run Terminal Help
20251007-SE-PT-AU-NZ
EXPLORER
OPEN EDITORS
20251007-SE-PT-AU-NZ
Module9
LabExercises
blogging_mongodb
controllers
commentController.js
index.js
likeController.js
postController.js
userController.js
models
comment.js
index.js
like.js
post.js
user.js
node_modules
routes
commentRoutes.js
likeRoutes.js
postRoutes.js
userRoutes.js
.env
dbConnect.js
LabExercise2.js
package-lock.json
package.json
server.js
calculator-app
LabExercise3
LabExercise4
LabExercise5
Module10
node_modules
.gitignore
package-lock.json
package.json
README.md
students.js
OUTLINE
TIMELINE
server.js U
New Request
GET
http://localhost:8080/api/comments/
Send
Query
Headers 2
Auth
Body 1
Tests
Pre Run
JSON
XML
Text
Form
Form-encode
GraphQL
Binary
JSON Content
Format
{
"user_id": "69b4d46a0b941ec3b4e9bcab",
"post_id": "69b4b0cba41d896ade2e77d7",
"content": "This is informative, jeremydelapenadev!"
}
Status: 200 OK
Size: 445 Bytes
Time: 4 ms
Response
Headers 6
Cookies
Results
Docs
{
"result": 200,
"data": [
{
"_id": "69b4b221a41d896ade2e77d9",
"content": "Congrats for successfully updating your first post!",
"post_id": "69b4b17ba41d896ade2e77d8",
"user_id": "69b4b0cba41d896ade2e77d7",
"created_at": "2026-03-14T00:56:00.998Z"
},
{
"_id": "69b4e7178915b6c84d919eca",
"content": "This is informative, jeremydelapenadev!",
"post_id": "69b4b0cba41d896ade2e77d7",
"user_id": "69b4d46a0b941ec3b4e9bcab",
"created_at": "2026-03-14T04:41:59.206Z",
"__v": 0
}
]
}
Response
Chart
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
}
{
user_id: '69b4d46a0b941ec3b4e9bcab',
post_id: '69b4b0cba41d896ade2e77d7',
content: 'This is informative, jeremydelapenadev!'
}
powersh...
main*
0 0
Go Live